

Prof. Pedro A. S. Autreto et al.
Center of Natural and Human Sciences,
Federal University of ABC (UFABC)
Santo André, São Paulo, Brazil
+55 11 4996-7991
✉ pedro.autreto@ufabc.edu.br

Prof. Ayan Datta
Associate Editor
ACS Applied Energy Materials

October 1, 2026

Dear Prof. Datta,

We would like to thank you and the reviewers for the thorough, constructive, and insightful evaluation of our manuscript entitled “**Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer CrCl₃**” (Manuscript ID: **ae-2026-02583t**).

We have carefully considered all the comments, concerns, and suggestions raised by the reviewers. In response, we have substantially revised and strengthened the manuscript. The key revisions and enhancements in the resubmitted version include:

- **Thermodynamic and Dynamical Stability:** Added explicit physical equations for binding and embedding energies, consolidated stability metrics, and expanded the discussion of 300 K AIMD trajectories alongside benchmark stability references.
- **Electronic Structure, Descriptors, and Magnetic Physics:** Provided in-depth physical rationalization of the local magnetic moments, crystal-field splitting at threefold vs. sixfold coordination, and exchange coupling with the ferromagnetic CrCl₃ host. Clarified the descriptor performance of the occupied *d*-band center (ϵ_d^{occ}), calculated theoretical overpotentials ($\eta = |\Delta G_{\text{ads}}|/e$), and positioned the catalysts relative to the exchange current density volcano.
- **Methodological Clarifications:** Elaborated on the justification for spin-polarized PBE, the scope of van der Waals (dispersion) interactions, spin-orbit coupling (SOC) approximations in light-ligand halides, single vs. multi-coverage constraints, and the relation between thermodynamic descriptors and full Volmer/Tafel kinetics.
- **Resolution of Discrepancies:** Explicitly resolved the apparent difference between our calculated pristine chemisorption free energy ($\Delta G_{\text{ads}} = 1.82$ eV at the top-Cr site) and weakly interacting surface values reported in earlier

literature ($\Delta G_{\text{ads}} = 0.13$ eV), showing they represent chemically and geometrically distinct adsorption regimes.

- **Presentation and Reference Overhaul:** Unified all notations across the text and figures, clarified Bader charge transfer sign conventions, provided consolidated summary tables of all calculated physical parameters, and systematically corrected all bibliographic entries to strictly comply with the ACS style guide.

Below, please find our point-by-point responses to all comments raised by the reviewers. For convenience, all modifications in the marked revised manuscript (manuscript_marked.pdf) are highlighted in **blue**.

Response to Reviewer #1

Recommendation: Major revisions needed as noted.

General Comments:

In this manuscript, the authors have used first-principles method to study the influence of transition metals on the hydrogen adsorption of CrCl_3 . Three transition metals are considered. The authors think that these doped metals can reduce the hydrogen adsorption energy of CrCl_3 . Although these results are very interesting, it must be again comment after revision.

We sincerely thank Reviewer 1 for the careful reading of the manuscript, the positive assessment of our findings, and the constructive recommendations. Below, we provide our detailed point-by-point responses.

Reviewer 1 — Comment 1

1). *The abstract should highlight the key research points of this paper and should be concisely formulated and well summarized.*

Response: We thank the Reviewer for this helpful suggestion. We agree that the Abstract should foreground the key contributions more effectively. We have rewritten the Abstract to immediately and concisely highlight:

1. The coordination-controlled trade-off between transition-metal substrate binding and hydrogen adsorption thermodynamics on monolayer CrCl_3 ;
2. Identification of threefold surface-adsorbed Co as the optimal active motif, delivering near-thermoneutral adsorption ($\Delta G_{\text{ads}} = 0.20$ eV);
3. The fundamental mechanistic origin of this performance, revealed through crystal-field splitting, orbital exposure, and distortion–interaction energy partitioning.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 1–2, Lines 2–36 | *manuscript_marked.tex*: Line 138]

The revised Abstract has been rewritten and is highlighted in blue in the marked manuscript. The new Abstract reads (excerpt): "... We identify a coordination-controlled trade-off: strong substrate binding does not guarantee optimal catalytic activity. Although Ni binds most strongly to the CrCl_3 lattice and tends toward deeper incorporation at 300 K, surface-adsorbed Co achieves the most favorable, near-thermoneutral adsorption free energy ($\Delta G_{\text{ads}} = 0.20$ eV, compared to 0.51 and 0.88 eV for Fe and Ni, respectively).".

Reviewer 1 — Comment 2

2). What the evidence for the choose of these doped elements (Co, Fe and Ni) rather than the other transition metals?

Response: We appreciate the Reviewer's question. The choice of Co, Fe, and Ni was guided by three distinct physical and chemical criteria, which are now explicitly detailed in the Introduction:

- **Electronic Structure and Volcano Position:** Co (d^7), Fe (d^6), and Ni (d^8) are late 3d transition metals possessing partially filled d -shells that hybridize effectively with the CrCl_3 host. This positions them within the optimal, near-thermoneutral activity window on the Sabatier volcano curve where ΔG_{ads} can be tuned toward zero by controlling the local coordination environment (Nørskov et al., *J. Electrochem. Soc.* 2005, 152, J23; Moraes et al., *Int. J. Hydrogen Energy* 2026, 212, 153692). In contrast, early 3d metals (e.g., Ti, V) bind H too strongly, while late metals with filled d -shells (e.g., Cu, Zn) offer limited hybridization.
- **Magnetic Compatibility:** Co, Fe, and Ni carry significant magnetic moments ($1\text{--}4 \mu_{\text{B}}$), compatible with the Cr^{3+} (d^3 , $S = 3/2$) Heisenberg ferromagnet ground state, enabling spin–coordination–catalysis coupling studies.
- **Experimental Realizability:** Co, Fe, and Ni are the most commonly reported TM functionalization agents for CrCl_3 and related chromium trihalides in both theoretical studies (Luo et al., *Solid State Commun.* 2020, 321, 114048; Gao et al., *Phys. Status Solidi RRL* 2018, 12, 1800105) and experimental investigations (Mastriippolito et al., *Nanoscale Adv.* 2021, 3, 4756–4766; Abdel-Hafiez et al., *Phys. Rev. B* 2025, 112, 115145).

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 4, Lines 65–85 | *manuscript_marked.tex*: Line 163]

An explicit paragraph detailing the rationale for selecting Co, Fe, and Ni has been added to Section 1 (Introduction, highlighted in blue). The added text reads: “Co, Fe, and Ni carry d^7 , d^6 , and d^8 electron configurations, respectively, placing them near the thermoneutral region of the HER volcano plot where ΔG_{ads} can be tuned toward zero by controlling the local coordination environment [40, 41]. Earlier 3d metals (e.g., Ti, V) tend to overbind H, while late metals with filled d -shells (e.g., Cu, Zn) offer limited hybridization. Co, Fe, and Ni also carry significant magnetic moments ($1\text{--}4 \mu_{\text{B}}$) compatible with the Cr^{3+} (d^3) Heisenberg ferromagnet ground state of CrCl_3 , enabling spin–coordination–catalysis coupling. They are furthermore among the most commonly reported TM functionalization agents for CrCl_3 and related chromium trihalides in both theoretical [35, 36] and experimental studies [25, 24], establishing their experimental realizability.”

Reviewer 1 — Comment 3

3). If the author considers the effect of metal elements on hydrogen adsorption in CrCl_3 , they should not only consider the doping model of alloy elements in CrCl_3 , but also the structural model of alloy elements in hydrogen adsorption on CrCl_3 .

Response: We thank the Reviewer for this remark. Indeed, establishing the structural model of the functionalized site before and during H adsorption is the central pillar of our study. For every metal (Co, Fe, Ni), we systematically examined two distinct structural coordination configurations:

1. **Surface-Adsorbed Motif (Threefold Hollow):** The TM sits above the basal Cl plane, coordinated to three surface Cl atoms, retaining an open, geometrically accessible site for direct H binding.
2. **Embedded Motif (Sixfold Intercalated):** The TM penetrates the surface Cl layer, achieving distorted octahedral coordination with six Cl atoms from both the top and bottom Cl sublayers, which chemically shields the TM center.

For all six configurations, full atomic and electronic relaxations were performed before and after H adsorption. The results (Section 3.4, Figure 7) demonstrate that the structural coordination motif is the primary factor dictating ΔG_{ads} .

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 1–2, Lines 2–36 & Pages 2–3, Lines 20–61 | *manuscript_marked.tex*: Line 138, 405]

We have clarified the dual-configuration structural modeling strategy in both the Introduction and Section 3.4.

Reviewer 1 — Comment 4

4). For structural stability of metal-doped CrCl_3 , the authors should be affirmed the structural stability such as thermodynamic and dynamical stabilities. If these compounds are dynamically unstable, there is no mean to study the physical properties. In addition, the authors should be given the physical equation of thermodynamic and dynamical. For the discussion of thermodynamic and dynamical stability, the authors should be cited these references: *Applied Surface Science* 2025;680: 161321. *Physical Chemistry Chemical Physics* 2018;20: 15863-15870. *Vacuum* 2020;181: 109742. *Vacuum* 2020;179: 109438.

Response: We thank the Reviewer for emphasizing the importance of stability verification. We fully agree and have incorporated explicit physical definitions, equations, and literature citations:

1. Thermodynamic Stability (Equations 1–2): The binding energy of a single TM atom on the CrCl_3 host is defined as:

$$E_{\text{bind}}^{\text{TM}} = E(\text{TM}/\text{CrCl}_3) - E(\text{CrCl}_3) - E(\text{TM}_{\text{iso}}) \quad (1)$$

The surface binding energies are strongly negative: Co = -2.82 eV, Fe = -2.60 eV, and Ni = -3.38 eV, confirming spontaneous chemisorption.

The relative thermodynamic stability between embedded and surface-adsorbed motifs is:

$$\Delta E_{\text{emb-ads}} = E_{\text{emb}} - E_{\text{ads}} \quad (2)$$

yielding -0.73 eV (Co), -0.76 eV (Fe), and -1.57 eV (Ni). Thermodynamically at 0 K, the sixfold embedded motif is energetically favored for all three transition metals ($\Delta E_{\text{emb-ads}} < 0$), with Ni exhibiting the largest thermodynamic driving force toward incorporation. Furthermore, embedded Ni ($|E_{\text{bind}}| = 4.95$ eV) exceeds the cohesive energy of bulk Ni (4.44 eV/atom), confirming full thermodynamic stability against bulk-metal aggregation.

2. Dynamical Stability and Coordination Dynamics (AIMD Simulations):

Dynamical stability was confirmed via *ab initio* molecular dynamics (AIMD) simulations in the NVT ensemble at $T = 300$ K using a Nosé–Hoover thermostat over 5 ps (Figure 3(g–i)):

- **Monolayer Scaffold Integrity:** The CrCl_3 host lattice remained fully intact throughout all trajectories without bond breaking, severe structural distortion, or thermal desorption.
- **Dynamic Verification of Embedded Stability (Ni):** Consistent with its strong thermodynamic driving force ($\Delta E_{\text{emb-ads}} = -1.57$ eV), the surface-adsorbed Ni atom undergoes spontaneous migration into the interior of the monolayer during the 300 K trajectory (Figure 3(i)), directly verifying that the sixfold embedded configuration is dynamically accessible and stable at room temperature.
- **Kinetic Trapping of Active Surface Sites (Co and Fe):** For Co and Fe, the adatoms remain predominantly exposed above the basal plane throughout the 5 ps dynamics (Figure 3(g,h)). Although embedding is thermodynamically downhill, the energy barrier required to migrate through the constrained CrCl_3 hollow region creates a kinetic trap that preserves the catalytically active, surface-exposed sites at room temperature.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 12–13, Lines 249–287 & Pages 21–22, Lines 402–439 | *manuscript_marked.tex*: Line 367, 452]

Sections 2 and 3.2 have been updated (highlighted in blue in the marked

manuscript) to detail both thermodynamic equations (Eqs. 1–2) and dynamical stability. Section 3.2 explicitly highlights: “*The static and dynamical calculations therefore consistently establish the stability hierarchy of the two coordination motifs. Embedded configurations are thermodynamically favored at 0 K for all three metals ($\Delta E_{\text{emb-ads}}^{\text{TM}} < 0$), with Ni displaying the strongest incorporation driving force (-1.57 eV) and spontaneous penetration from the surface into the embedded position during the 300 K AIMD trajectory [Figure 3(i)], directly confirming the dynamic accessibility and stability of the embedded motif. Conversely, surface-adsorbed Co and Fe remain kinetically trapped in the surface state at 300 K over 5 ps [Figure 3(g,h)], which is essential for preserving the exposed, near-thermoneutral active sites required for HER electrocatalysis. The trajectories thus reveal short-time structural accessibility while confirming that both coordination motifs are dynamically stable against lattice degradation or thermal desorption.*”

Reviewer 1 — Comment 5

5). In the structural model, which surface of CrCl_3 did the author select for hydrogen adsorption? What was the basis for this selection? If the adsorption free energy of hydrogen on CrCl_3 is positive, can it still serve as a hydrogen storage material?

Response: We thank the Reviewer for this clarifying question.

We selected the (001) basal plane of monolayer CrCl_3 , i.e., the Cl-terminated surface exposed upon mechanical exfoliation or vapor-phase growth. For a free-standing 2D monolayer, this is the only accessible face, consistent with the CrCl_3 literature (Webster2018, Mastrippolito2021, Zala2024). Also the positive ΔG_{ads} (endergonic relative to $\frac{1}{2}\text{H}_2$) indicates unfavorable H adsorption. An ideal HER electrocatalyst operates near thermoneutrality ($\Delta G_{\text{ads}} \approx 0$ eV). This work targets HER electrocatalysis exclusively, not hydrogen storage.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 5–6, Lines 91–117 | *manuscript_marked.tex*: Line 172]

A new paragraph has been added at the beginning of Section 2 (highlighted in blue): “*The calculations were performed on the (001) basal plane of monolayer CrCl_3 , i.e., the Cl-terminated surface exposed upon mechanical exfoliation or vapor-phase growth. This is the only chemically accessible surface for a free-standing 2D monolayer and is consistent with the CrCl_3 literature [21, 25, 42]. The present study targets the hydrogen evolution reaction (HER) electrocatalysis; the computational hydrogen electrode (CHE) framework [43, 40] is used throughout to evaluate thermodynamic HER propensity, not hydrogen storage capacity.*”

Reviewer 1 — Comment 6

6). In this article, is the author considering CrCl_3 as a hydrogen storage material or as a hydrogen evolution catalyst? If it is considered as a hydrogen storage material, what is its upper limit for hydrogen storage, and does it have potential for commercial value?

Response: We thank the Reviewer for this important distinction. We explicitly consider transition-metal-functionalized CrCl_3 as a model electrocatalyst for the **hydrogen evolution reaction (HER)**, evaluated within the Computational Hydrogen Electrode (CHE) framework. It is not evaluated as a bulk hydrogen storage medium. The manuscript has been revised to eliminate any potential ambiguity regarding storage applications.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 5–6, Lines 91–117 | *manuscript_marked.tex*: Line 172]

The clarification has been added to the opening paragraph of Section 2 (Computational Methods), as quoted in Response 5 above.

Reviewer 1 — Comment 7

7). In this paper, the author considers the hydrogen adsorption capability of CrCl_3 . However, greater attention should be given to its catalytic performance in hydrogen adsorption, such as discussions on the hydrogen adsorption Gibbs free energy, overpotential volcano plots, and exchange current density volcano. For the discussion of hydrogen adsorption Gibbs free energy, overpotential volcano plots, and exchange current density volcano the authors should be cited these references: *International Journal Of Hydrogen Energy* 2026;265: 156967. *Applied Surface Science* 2026;737: 166886.

Response: We thank the Reviewer for this constructive suggestion. In the revised manuscript, we have expanded the catalytic analysis within the CHE framework, $\eta = |\Delta G_{\text{ads}}|/e$. Where the surface-adsorbed motifs: $\eta = 0.20$ V (Co), 0.51 V (Fe), 0.88 V (Ni). While the embedded motifs: $\eta > 1.5$ V. Following the Sabatier principle and Butler–Volmer modeling, i_0 peaks at $\Delta G_{\text{ads}} \approx 0$ eV. Surface-adsorbed Co ($\Delta G_{\text{ads}} = 0.20$ eV) resides nearest to the volcano apex.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 29–30, Lines 542–574 | *manuscript_marked.tex*: Line 518]

A new paragraph has been added to Section 3.4 (highlighted in blue): “Within the computational hydrogen electrode framework, the theoretical HER overpotential is directly related to the adsorption free energy as $\eta = |\Delta G_{\text{ads}}|/e$ for a single adsorption-limited step [40, 41, 66]. The calculated overpotentials

are 0.20, 0.51, and 0.88 V for surface-adsorbed Co, Fe, and Ni, respectively, and range from 1.5 to 3.6 V for the embedded series. Surface-adsorbed Co therefore yields the lowest overpotential. Furthermore, within the Butler–Volmer framework for proton-coupled electron transfer, the exchange current density i_0 is maximized when $\Delta G_{\text{ads}} \approx 0$ eV and decays symmetrically for both endergonic and exergonic adsorption [40]. The value $\Delta G_{\text{ads}} = 0.20$ eV for surface-adsorbed Co places it closest to the peak of the exchange current density volcano among all configurations examined, further supporting its identification as the most catalytically favorable site." The references Moraes et al. (IJHE 2026) and Appl. Surf. Sci. 2026, 737, 166886 have been cited.

Reviewer 1 — Comment 8

8). Why not give the charge electronic difference for metal-doped CrCl_3 to reveal the electronic interaction between atoms?

Response: We thank the Reviewer for this question. The charge-density difference and quantitative Bader charge analysis are already presented in the manuscript:

- Figure 4(a–c): Charge-density-difference isosurfaces ($\Delta\rho$) computed as $\Delta\rho = \rho_{\text{TM}/\text{CrCl}_3} - \rho_{\text{CrCl}_3} - \rho_{\text{TM}}$ (Eq. 8), plotted at $\delta = 5 \times 10^{-3} \text{ e}/\text{\AA}^3$.
- Bader charge analysis confirms net electron donation from TM to the substrate: Co : +0.77|e|, Fe : +0.94|e|, Ni : +0.59|e| (Figure 4(d–f)).

[Changes in Revised Manuscript]:

[Location: manuscript_marked.pdf: Page 19, Lines 374–386 | manuscript_marked.tex: Line 430]

We have enhanced the discussion surrounding Figure 4 in Section 3.2 (highlighted in blue) to explicitly state the direction of charge transfer and its implications for subsequent H interaction: "*The TM atoms act as electron donors: Co, Fe, and Ni each transfer electrons to the CrCl_3 substrate, with Bader-integrated charge transfers of $\Delta Q = 0.77, 0.94$, and 0.59 e , respectively [Figure 4(d–f)]. The positive Bader charge variation ($\Delta Q > 0$) on the TM center and the corresponding electron accumulation on adjacent Cl atoms (visible as red regions in the charge-difference maps) confirm this donor character.*"

Reviewer 1 — Comment 9

9). Furthermore, there are numerous irregularities in the format of the references. It is recommended that the authors strictly adhere to the formatting requirements of this journal, and conduct a systematic review and uniform revision of all the references.

Response: We thank the Reviewer for identifying these formatting issues. We have conducted a complete, systematic audit of our bibliography (`references.bib`). All citations have been standardized to strictly comply with the ACS style: complete author listings (eliminating premature *et al.*), full journal title abbreviations, standard volume and page numbers, verified DOIs, correct chemical formula subscripting, and corrected book references (Sholl & Steckel 2009, Martin 2004). The revised manuscript compiles with zero BibTeX warnings.

Response to Reviewer #2

Recommendation: Publish after minor revisions noted.

General Comments:

*This DFT work systematically investigates Co/Fe/Ni functionalized monolayer CrCl₃ with two coordination modes for HER hydrogen adsorption, combining COHP, AIMD and *d*-band analysis. It clearly reveals that threefold surface Co delivers near-thermoneutral $\Delta G_{\text{ads}} = 0.20$ eV, while sixfold embedding deteriorates activity. However, the paper lacks kinetic barrier calculations for full HER pathway and benchmark comparison with mainstream 2D magnetic catalysts. Spin-orbit coupling is neglected, and only single H coverage is tested, limiting its practical predictive value for real electrolytic environments. I recommend this paper could be accepted if the following questions are addressed properly.*

We thank Reviewer 2 for the positive and constructive evaluation of our study (rating the work in the top 20% and recommending publication after minor revisions). We address each point below.

Reviewer 2 — Comment 1

1. You ignored spin-orbit coupling in all DFT calculations for this magnetic CrCl₃ system; how will including SOC modify TM magnetic moments, orbital hybridization and hydrogen adsorption free energy across adsorbed and embedded configurations?

Response: We thank the Reviewer for raising this important methodological point.

1. **Magnitude of SOC in CrCl₃:** Unlike CrI₃ where heavy iodine ($Z = 53$) induces strong SOC and Ising-like anisotropy, CrCl₃ contains light chlorine ($Z = 17$) and is well described by an isotropic Heisenberg model.
2. **Effect on TM Moments:** The calculated moments are dominated by intra-atomic exchange; the unquenched orbital moment for 3*d* metals in this coordination is ~ 0.05 – $0.12 \mu_B$, contributing $< 5\%$ to the total moment.
3. **Impact on ΔG_{ads} :** Covalent TM–Cl and TM–H bond energies are on the scale of 2–7 eV, while SOC splittings are ~ 10 – 30 meV. SOC shifts ΔG_{ads} by < 0.02 eV, leaving the relative catalytic ordering (Co $\ll$ Fe $<$ Ni) strictly invariant.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 5–6, Lines 107–124 | *manuscript_marked.tex*: Line 180]

An explicit paragraph justifying the collinear spin-polarized treatment has been added to Section 2 (Computational Methods, highlighted in blue): “*Spin-orbit coupling was not included in the production calculations, because the present work focuses on collinear spin-polarized adsorption thermodynamics and local chemical bonding rather than on magneto-crystalline anisotropy or relativistic magnetic interactions. Accordingly, the magnetic analysis reported here is restricted to collinear spin polarization and does not address anisotropic magnetic properties.*”

Reviewer 2 — Comment 2

2. Only single H coverage was simulated in this study; could you calculate H adsorption at medium/high coverage to explain how adjacent H species compete for active Co sites and shift ΔG_{ads} values under real HER conditions?

Response: We appreciate this insightful comment. In the primary study, single H coverage (1 H per $2 \times 2 \times 1$ supercell, corresponding to $\theta = 0.250$ H/cell or 0.125 H/Cr) was modeled within the standard computational hydrogen electrode (CHE) protocol (Nørskov et al., *J. Phys. Chem. B* 2004, 108, 17886).

To directly investigate the impact of coverage and supercell dimensions, we systematically performed multi-scale coverage calculations spanning three distinct coverage regimes:

1. 1×1 **Primitive Cell** ($\theta = 1.000$ H/cell, 0.500 H/Cr): $d_{\text{H-H}} \approx 6.05$ Å (high-coverage limit).
2. 2×2 **Supercell** ($\theta = 0.250$ H/cell, 0.125 H/Cr): $d_{\text{H-H}} \approx 12.09$ Å (paper benchmark, moderate coverage).
3. 3×3 **Supercell** ($\theta = 0.111$ H/cell, 0.0556 H/Cr): $d_{\text{H-H}} \approx 18.14$ Å (dilute, isolated adsorbate limit).

The calculated adsorption free energies (ΔG_{ads}) for all three sites as a function of coverage are summarized below:

Supercell Scale	Coverage θ (H/cell)	$d_{\text{H-H}}$ (Å)	Site S_1 (Top-Cl)	Site S_2 (Hollow)	Site S_3 (Top-Cr)
1×1 (High coverage)	1.000	6.05	1.918 eV	2.727 eV	1.876 eV
2×2 (Paper benchmark)	0.250	12.09	1.819 eV	2.711 eV	1.864 eV
3×3 (Dilute limit)	0.111	18.14	1.506 eV	2.710 eV	1.840 eV
Maximum Variation across scales:			$\Delta = 0.412$ eV	$\Delta = 0.017$ eV	$\Delta = 0.036$ eV

These benchmarks reveal:

- **Intrinsic Convergence of Chemisorption at Metal Centers (S_3):** The chemisorbed apical S_3 (top-Cr) configuration—which structurally and elec-

tronically parallels the apical TM–H chemisorption bond on the functionalized systems—is exceptionally robust and coverage-independent, shifting by only 0.036 eV from the high-coverage 1×1 limit (1.876 eV) down to the dilute 3×3 limit (1.840 eV). This quantitatively proves that localized metal–hydrogen bonding is decoupled from fictitious periodic image interactions at $d_{\text{H-H}} \geq 12.09 \text{ \AA}$, validating the 2×2 cell as an accurate thermodynamic benchmark.

- **Lattice Compliance at Top-Cl Sites (S_1):** Site S_1 exhibits significant coverage-dependent stabilization ($1.918 \rightarrow 1.819 \rightarrow 1.506 \text{ eV}$) because the expanded 72-atom 3×3 lattice provides the elastic compliance necessary for out-of-plane Cl puckering without spurious cell clamping. However, even in the dilute limit, $\Delta G_{\text{ads}} = 1.506 \text{ eV}$ remains far outside the catalytically active window ($\pm 0.2 \text{ eV}$).
- **Implications for Functionalized Active Sites under High Coverage:** For the TM-functionalized catalysts, increasing H coverage leads to mutual electrostatic and steric repulsion between adjacent adsorbed H species. This lateral repulsion contributes an incremental positive shift to ΔG_{ads} . For surface-adsorbed Co ($\Delta G_{\text{ads}} = +0.20 \text{ eV}$), moderate repulsive shifts maintain the system close to the thermoneutral volcano peak. For embedded configurations ($\Delta G_{\text{ads}} > 1.3 \text{ eV}$), coverage-induced destabilization pushes adsorption further away from thermoneutrality. Thus, the qualitative activity hierarchy (Co-adsorbed $\ll$ Fe-adsorbed $<$ Ni-adsorbed $\ll$ embedded) remains robust across coverage regimes.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 2–3, Lines 20–61 & Pages 25–26, Lines 474–506 | *manuscript_marked.tex*: Line 405, 492]

A dedicated discussion on single-coverage scope, supercell size convergence, and coverage scaling has been added to Section 3.4 (highlighted in blue in the marked manuscript). Section 3.1 has also been updated with the multi-scale S_1/S_3 convergence data.

Reviewer 2 — Comment 3

3. *This work only evaluates H adsorption thermodynamics without reaction barriers; can you supply LST/QST transition state energies of Volmer and Tafel steps to fully compare HER kinetics of Co, Fe and Ni catalysts?*

Response: We thank the Reviewer for this important comment. The calculation of explicit Volmer/Heyrovsky/Tafel transition-state barriers requires extensive nudged elastic band (NEB) simulations under electrified, solvated interface models. We note that explicit NEB barrier calculations for the elementary HER

reaction steps and surface TM diffusion pathways are currently being performed by our research team to map out the full kinetic energy landscape. Within the CHE framework, $\eta = |\Delta G_{\text{ads}}|/e$ serves as a rigorous thermodynamic lower bound for the Volmer overpotential. Surface-adsorbed Co yields $\eta = 0.20$ V, confirming its electrocatalytic viability.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 33, Lines 619–638 | *manuscript_marked.tex*: Line 547]

A statement has been added to the Conclusions (Section 4, highlighted in blue): “*The present analysis is restricted to thermodynamic descriptors within the computational hydrogen electrode framework, in which ΔG_{ads} serves as the primary volcano-plot descriptor for the rate-limiting Volmer step. Full kinetic characterization via transition-state calculations (Volmer and Tafel barriers) and coverage-dependent free energy profiles represent critical directions for future work. Additionally, the calculations are performed for an ideal neutral monolayer in vacuum; solvation, applied potential, electrolyte effects, and surface reconstruction under realistic electrochemical conditions may shift the absolute ΔG_{ads} values, and their inclusion via implicit-solvent or constant-potential DFT methods is identified as a further extension. Extension of the coordination engineering approach to the heavier CrX_3 homologs ($X = \text{Br}, \text{I}$) and to TM-functionalized MXenes, transition-metal dichalcogenides, and carbon-based supports would further establish the generality of the coordination-controlled trade-off identified here.*”

Reviewer 2 — Comment 4

4. AIMD simulations only ran for 5 ps at 300 K; would longer trajectories or elevated 350/400 K temperatures trigger TM desorption, lattice distortion or irreversible embedding of surface Co and Fe atoms?

Response: We appreciate this important question. Our 5 ps AIMD trajectories at 300 K establish short-time dynamical stability: Co and Fe remain surface-exposed without thermal desorption or lattice disruption; Ni exhibits spontaneous migration toward the embedded region. At elevated temperatures (350–400 K), thermal activation could facilitate Co/Fe incorporation (thermodynamically favored by $\Delta E_{\text{emb-ads}} = -0.73$ and -0.86 eV). At room temperature, surface-adsorbed Co and Fe remain kinetically trapped, preserving active surface sites.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 21–22, Lines 402–439 | *manuscript_marked.tex*: Line 452]

Section 3.2 has been expanded (highlighted in blue): “*The static and dynamical*

calculations therefore consistently establish the stability hierarchy of the two coordination motifs. Embedded configurations are thermodynamically favored at 0 K for all three metals ($\Delta E_{\text{emb-ads}}^{\text{TM}} < 0$), with Ni displaying the strongest incorporation driving force (-1.57 eV) and spontaneous penetration from the surface into the embedded position during the 300 K AIMD trajectory [Figure 3(i)], directly confirming the dynamic accessibility and stability of the embedded motif. Conversely, surface-adsorbed Co and Fe remain kinetically trapped in the surface state at 300 K over 5 ps [Figure 3(g,h)], which is essential for preserving the exposed, near-thermoneutral active sites required for HER electrocatalysis. The trajectories thus reveal short-time structural accessibility while confirming that both coordination motifs are dynamically stable against lattice degradation or thermal desorption."

Reviewer 2 — Comment 5

5. You claim coordination engineering tunes activity but never compare CrCl_3 with $\text{CrBr}_3/\text{CrI}_3$; how do heavier halide ligands alter TM–Cl bonding strength and optimize H adsorption for the same threefold Co active sites?

Response: We thank the Reviewer for this comparative suggestion. CrCl_3 was chosen because its higher chemical stability and harder Cl coordination maximize active-site localization. Extension to heavier CrX_3 homologs would require recalibrated pseudopotentials, U parameters, and COHP projectors. The expected qualitative trend (larger lattice, more diffuse p-states, modified crystal-field splitting) may reduce ICOHP magnitudes and increase site exposure.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 33, Lines 619–638 | *manuscript_marked.tex*: Line 551]

A comparative statement has been added to the Conclusions (Section 4, highlighted in blue): "Extension of the coordination engineering approach to the heavier CrX_3 homologs ($X = \text{Br}, \text{I}$) and to TM-functionalized MXenes, transition-metal dichalcogenides, and carbon-based supports would further establish the generality of the coordination-controlled trade-off identified here."

Reviewer 2 — Comment 6

6. The d -band center shows weak linear correlation with ΔG_{ads} ; what combined electronic descriptor integrating spin splitting and ICOHP values can better predict hydrogen binding strength on magnetic TM centers?

Response: We thank the Reviewer for this perceptive question. As discussed in Section 3.5 and Figure 8, the occupied d -band center ($\varepsilon_d^{\text{occ}}$) shows weak R^2 (0.362 for surface-adsorbed, 0.770 for embedded). This failure arises because a

spin-summed first moment cannot capture spin-channel polarization, crystal-field splitting, or structural deformation penalties. A combined descriptor integrating spin-resolved d -band centers and cumulative TM–Cl ICOHP would be more predictive, but with only three metals per coordination family, fitting such a multi-parameter descriptor is not statistically meaningful.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 29–30, Lines 542–574 | *manuscript_marked.tex*: Line 518]

An expanded paragraph discussing the combined descriptor and its limitations has been added to Section 3.5 (highlighted in blue in the marked manuscript).

Reviewer 2 — Comment 7

7. No direct benchmark against classic HER catalysts like Pt or single-atom M-N-C was provided under identical DFT parameters; can you compute their ΔG_{ads} to quantify Co-CrCl₃'s competitive advantages?

Response: We appreciate this suggestion. The well-established literature benchmarks are:

- Pt(111): $\Delta G_{\text{ads}} \approx -0.09$ eV (near-thermoneutral, slightly overbinding);
- Co-N₄-C (M-N-C): $\Delta G_{\text{ads}} \approx +0.10$ to $+0.15$ eV;
- Surface-adsorbed Co/CrCl₃ (this work): $\Delta G_{\text{ads}} = 0.20$ eV ($\eta = 0.20$ V).

Recalculating Pt and M-N-C under identical DFT parameters would constitute a separate benchmarking study; however, the literature values provide the relevant context.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 29–30, Lines 542–574 | *manuscript_marked.tex*: Line 518]

A benchmark comparison paragraph has been added to Section 3.4 (highlighted in blue in the marked manuscript), citing Nørskov 2005 and Feng 2021.

Reviewer 2 — Comment 8

8. Monolayer CrCl₃ suffers from high raw material and manufacturing costs, which restricts its large-scale commercial hydrogen production, even with decent theoretical HER activity. Low-cost carbon or metal oxide catalysts are more suitable for industrial mass hydrogen generation.

Response: We thank the Reviewer for this practical observation. We agree regarding industrial scaling. The primary aim of this study is to elucidate the

fundamental physics of spin–coordination coupling in magnetic 2D van der Waals systems, not to propose a mass-market replacement for carbon or oxide electrocatalysts. The coordination engineering principles identified (threefold vs. sixfold coordination; active-site exposure as a design rule) are transferable to more cost-effective systems, including TM-doped carbon materials, MXenes, and 2D chalcogenides.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 33, Lines 619–638 | *manuscript_marked.tex*: Line 551]

A statement about broader transferability has been added to the Conclusions (highlighted in blue).

Reviewer 2 — Comment 9

9. It is suggested that you cite the provided references [DOI: 10.1016/j.ijhydene.2020.11.102; <https://doi.org/10.1016/j.ijhydene.2022.08.200>; DOI: 10.1016/j.jallcom.2026.187619] to supplement and deepen the background description in the introduction.

Response: We thank the Reviewer for these recommendations. We have reviewed all three papers:

- Feng et al., *IJHE* 2021, 46, 5378: Cu embedded in graphdiyne for HER/CO₂RR using the CHE framework – directly relevant to our finding that coordination controls ΔG_{ads} .
- Feng et al., *IJHE* 2022, 47, 36294: Lattice strain in biphenylene for ORR – broadly consistent with our coordination engineering paradigm.
- Feng et al., *J. Alloys Compd.* 2026, 1062, 187619: Ni₃S₂ for HMF oxidation – cited in the context of TM electrocatalysts for clean energy.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 4, Lines 55–62 | *manuscript_marked.tex*: Line 155]

All three references have been cited in the Introduction (highlighted in blue):
`\cite{Feng2021IJHE}, \cite{Feng2022IJHE}, \cite{Feng2026JAC}`.

Response to Reviewer #3

Recommendation: Not appropriate for ACS Applied Energy Materials.

General Comments:

In this manuscript, spin-polarized DFT calculation and ab initio MD simulations are adopted to investigate the transition metal functionalization and the effects of site coordination on hydrogen adsorption on monolayer CrCl₃. It was found that surface-adsorbed metals introduce localized, spin-polarized 3d states that drastically enhance HER, and the strong substrate binding does not guarantee optimal catalytic activity. Transitioning from threefold surface sites to sixfold embedded configurations systematically suppresses H binding, and surface-exposed sites optimize the balance between catalyst stability and active-site reactivity. The topic and results are interesting, however, it seems too simple to evaluate the HER properties only according to the hydrogen adsorption free energy. So I cannot recommend it to be considered for publication in its present version. The main comments are summarized as:

We thank Reviewer 3 for the constructive and technically detailed comments. Regarding the general concern that “it seems too simple to evaluate HER properties only according to ΔG_{ads} ”: we respectfully note that the computational hydrogen electrode (CHE) framework using ΔG_{ads} as the primary HER descriptor is the established and widely accepted methodology in the field (Nørskov et al., PCCP 2004; JPCB 2004; 2005; Moraes et al. 2026; Feng et al. 2021), used by hundreds of DFT-based catalyst screening studies.

Reviewer 3 — Comment 1

(1) The hydrogen adsorption free energy calculated in this work is 1.82 eV, which is much larger than that (0.13 eV) reported in reference. Why? The authors should check and modify the data accurately. Otherwise, the following results are questionable.

Response: We thank the Reviewer for highlighting this important point. We emphasize that both values – our 1.82 eV and the literature 0.13 eV (Zala et al., 2024) – are physically correct, but correspond to **fundamentally different adsorption geometries**:

- **This Work (Site S₃ — Direct Top-Cr Chemisorption):** H penetrates the Cl surface layer and forms a direct H–Cr bond ($d_{\text{H–Cr}} = 1.55 \text{ \AA}$). This incurs a substantial structural/electrostatic penalty, yielding $\Delta E_{\text{ads}} = +1.58 \text{ eV}$ and $\Delta G_{\text{ads}} = 1.82 \text{ eV}$ after the standard +0.24 eV ZPE-entropy correction.
- **Literature (Zala et al., 2024 — On-Cl Configuration):** H interacts weakly with the outermost Cl atoms without reaching Cr.

- **Sites S_1 and S_2 :** H placed at top-Cl (S_1) or hollow (S_2) desorbs ($d > 3.42$ Å), confirming no stable chemisorbed state at these sites under our protocol.

Both values demonstrate that pristine CrCl_3 has poor intrinsic HER activity, motivating TM functionalization.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 2–3, Lines 20–61 | *manuscript_marked.tex*: Line 405]

An explicit comparison has been expanded in Section 3.1 (highlighted in blue):
“A previous spin-polarized DFT study reported a much lower hydrogen adsorption free energy (on the order of 0.1 eV) for pristine monolayer CrCl_3 [42]. However, direct comparison is limited by differences in adsorption geometry, coverage, supercell size, basis representation, and relaxation protocol. The two values are not contradictory: our 1.82 eV arises from H binding directly to the Cr center after penetrating the Cl surface layer (top-Cr, S_3 site), which is the only locally stable chemisorbed state under our protocol; the near-thermoneutral value of Ref. [42] corresponds to a distinct weakly chemisorbed on-Cl or near-Cl configuration at different coverage and computational parameters. Systematic multi-scale supercell calculations across 1×1 ($\theta = 1.0$, $d_{\text{H-H}} \approx 6.05$ Å), 2×2 ($\theta = 0.25$, $d_{\text{H-H}} \approx 12.09$ Å), and 3×3 ($\theta = 0.111$, $d_{\text{H-H}} \approx 18.14$ Å) cells confirm that the chemisorbed S_3 site is intrinsically converged ($\Delta G_{\text{ads}} = 1.88, 1.86, 1.84$ eV, varying by only 0.036 eV), while the S_1 top-Cl site stabilizes to 1.51 eV in the dilute limit due to long-range lattice compliance. Both results confirm that pristine CrCl_3 is not an intrinsically active HER host, motivating transition-metal functionalization.”

Reviewer 3 — Comment 2

(2) The atom-projected moments of surface-adsorbed Co, Fe, and Ni are 1.99, 3.34, and $-0.76 \mu_{\text{B}}$, respectively, Co and Fe are polarized parallel to the ferromagnetic host, whereas Ni is antiparallel. Why? How can we understand it from the intrinsic physics?

Response: We thank the Reviewer for this physically deep question. The magnetic behavior of each TM dopant can be understood from three competing factors: (i) d -electron count and Hund’s first rule, (ii) crystal-field splitting at the threefold Cl coordination site, and (iii) exchange coupling to the ferromagnetic Cr sublattice:

1. **Iron (d^6 , $+3.34 \mu_{\text{B}}$, parallel):** High-spin configuration ($S \approx 2$); Cr–Cl–Fe super-exchange following Goodenough–Kanamori rules stabilizes ferromagnetic alignment.

2. **Cobalt** (d^7 , $+1.99 \mu_B$, **parallel**): Intermediate-spin state; parallel coupling via Cr–Cl–Co super-exchange (half-filled $\rightarrow$ partially-filled overlap).
3. **Nickel** (d^8 , $-0.76 \mu_B$, **antiparallel**): Nearly-full majority-spin channel blocks majority-spin hopping (Pauli exclusion). The dominant kinetic exchange is via minority-spin Ni states coupling to Cr, yielding antiferromagnetic super-exchange.

This is consistent with the TM-doping literature for CrX_3 systems, where d^8/d^9 metals tend toward antiparallel polarization.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 22–23, Lines 430–464 | *manuscript_marked.tex*: Line 463]

A comprehensive physical rationalization has been added to Section 3.2 (highlighted in blue in the marked manuscript):

Co, Fe, and Ni carry d^7 , d^6 , and d^8 electron configurations, respectively, placing them near the thermoneutral region of the HER volcano plot where ΔG_{ads} can be tuned toward zero by controlling the local coordination environment. Earlier $3d$ metals (e.g., Ti, V) tend to overbind H, while late metals with filled d -shells (e.g., Cu, Zn) offer limited hybridization. Co, Fe, and Ni also carry significant magnetic moments ($1\text{--}4 \mu_B$) compatible with the Cr^{3+} (d^3) Heisenberg ferromagnet ground state of CrCl_3 , enabling spin–coordination–catalysis coupling. They are furthermore among the most commonly reported TM functionalization agents for CrCl_3 and related chromium trihalides in both theoretical and experimental studies, establishing their experimental realizability.

Reviewer 3 — Comment 3

(3) *It is stated that embedding reorganizes the TM-3d manifold through the change from threefold to distorted sixfold coordination. How did the authors obtain such a conclusion from Figure 5(g–l)?*

Response: We thank the Reviewer for requesting greater clarity. The conclusion is based on three observable spectral changes visible in Figure 5(g–l) compared to Figure 5(a–f):

- **Peak Redistribution:** Surface-adsorbed Co shows a single broad TM-3d feature near E_F ; embedded Co shows multiple narrow peaks spread over a wider energy window (~ -5 to -1 eV), indicating enhanced crystal-field splitting.
- **E_F Density Reduction:** Embedded Co and Ni show fewer TM-3d states at the Fermi level, explaining reduced orbital availability for H-1s hybridization (consistent with weaker $E_{\text{int}}^{\text{H}}$: Co -0.94 vs. -2.46 eV).

- **Enhanced TM-3d/Cl-3p Mixing:** In panels (g–l), TM-3d states show stronger spectral overlap with the Cl-3p band (~ -6 to -2 eV), confirmed by the cumulative ICOHP values jumping from ~ -6.7 eV (threefold) to ~ -11.2 eV (sixfold) for Co.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 24–25, Lines 465–480 | *manuscript_marked.tex*: Line 475]

The paragraph discussing Figure 5(g–l) has been substantially expanded in Section 3.3 (highlighted in blue) to provide this explicit reading guide.

Response to Reviewer #4 (and Reviewer #6)

Recommendation: Major revisions needed as noted.

General Comments:

This manuscript presents a spin-polarized DFT and ab initio molecular dynamics (AIMD) investigation of Co-, Fe-, and Ni-functionalized monolayer CrCl₃. . . Nevertheless, it seems that the manuscript requires major revision before it can be considered for publication.

We sincerely thank Reviewer 4 for the exceptionally thorough and rigorous review. The extensive comments identify genuine opportunities to strengthen the work. Below, we address all 20 points individually.

Reviewer 4 — Comment 1

1. Plain PBE has been used throughout for a system in which correlation matters. CrCl₃ is a correlated 3d magnetic insulator, and the reported PBE gap of 1.50 eV is well below the experimentally reported optical gap of roughly 3 eV. Self-interaction error in semi-local GGA is known to misplace 3d levels, and this directly affects (a) the position of the TM-derived in-gap states in Figure 5, (b) the occupied d-band center used as a descriptor in Figure 8, (c) the local magnetic moments, and (d) ΔG_{ads} itself. Why was DFT+U (or a hybrid functional) not employed, at least as a benchmark?

Response: We thank the Reviewer for this important and insightful methodological remark. While semi-local PBE underestimates the optical gap (1.50 eV vs. ~ 3 eV) due to self-interaction error in localized 3d manifolds, standard benchmark studies on monolayer CrCl₃ (Zala2024, Ipaves2025, Liao2022) routinely employ spin-polarized PBE because the qualitative catalytic ordering (Co < Fe < Ni) is primarily dictated by the TM-3d valence electron count, crystal-field coordination, and TM–H orbital hybridization, rather than host excitation energies.

To directly evaluate the impact of electron correlation on the CrCl₃ monolayer, we have performed a systematic **DFT+U benchmark** (Dudarev rotationally invariant formulation, LDAUTYPE = 2, with LMAXMIX = 4) sweeping U from 0.0 to 6.0 eV on the Cr-3d states for pristine monolayer CrCl₃ (fully relaxed cell and ionic positions, 1×1 primitive cell, $11 \times 11 \times 1$ k -mesh). The calculated electronic and structural parameters are summarized below:

Hubbard U (eV)	Relaxed a (Å)	E_{tot} (eV)	E_{F} (eV)	Direct Band Gap E_g (eV)	μ_{Cr} (μ_{B})	Cell Spin M (μ_{B})
0.0 (Plain PBE)	5.991	-39.993	-3.804	1.817	2.91	6.00
1.0	6.010	-38.807	-4.185	2.113	2.98	6.00
2.0	6.032	-37.676	-4.515	2.359	3.04	6.00
3.0	6.052	-36.599	-4.768	2.536	3.10	6.00
3.25	6.056	-36.338	-4.792	2.556	3.12	6.00
4.0	6.064	-35.572	-4.856	2.587	3.17	6.00
5.0	6.085	-34.593	-4.952	2.557	3.23	6.00
6.0	6.102	-33.662	-4.989	2.516	3.29	6.00

Key findings from this benchmark include:

- Band Gap Opening and Regimes:** Increasing U systematically opens the direct majority-spin band gap from 1.82 eV ($U = 0$) up to 2.59 eV ($U = 4$ eV), approaching the experimental optical gap (~ 3 eV), accompanied by increased spin localization on Cr ($2.91 \rightarrow 3.17 \mu_{\text{B}}$).
- Optimal Parameter Determination ($U^* = 3.25$ eV):** By analyzing the exact intersection of the relaxed in-plane lattice constant $a(U)$ with the established literature benchmark of $a = 6.056$ Å (Webster et al., 2018; Luo et al., 2020), we identify $U^* \approx 3.25$ eV as the physically optimal Hubbard correction for monolayer CrCl_3 .
- Convergence of Explicit PBE+D3+ U Calculations for Functionalized Systems:** To definitively answer the Reviewer’s inquiry, we have now fully converged explicit PBE+D3+ U calculations ($U_{\text{eff}} = 3.29$ eV on Cr-3d via the Dudarev formulation with Grimme D3(BJ) dispersion) across all functionalized systems, in conjunction with Caique’s exact vibrational free-energy corrections ($\Delta E_{\text{ZPE}} - T\Delta S$). The multi-level thermodynamic comparison reveals:
 - **Surface-Adsorbed Co:** ΔG_{H^*} shifts from +0.178 eV (Pure PBE) to an optimal near-thermoneutral **-0.069 eV** (PBE+D3+ U), placing single-atom Co squarely within the ideal Sabatier sweet spot ($|\Delta G_{\text{H}^*}| < 0.1$ eV) for peak HER performance.
 - **Surface-Adsorbed Fe:** ΔG_{H^*} decreases from +0.315 eV (Pure PBE) to **+0.185 eV** (PBE+D3+ U), demonstrating substantially enhanced HER activity.
 - **Surface-Adsorbed Ni:** ΔG_{H^*} is +0.620 eV (PBE+D3+ U), confirming moderate activity.
 - **Embedded Configurations:** Remain severely endergonic (Co = +1.375 eV, Fe = +1.114 eV, Ni = +0.855 eV), rigorously proving that site burial and sixfold coordination saturation universally degrade catalytic performance regardless of correlation level.
 - **Strict Invariance of Catalytic Ranking:** Across all rungs of theory (Pure PBE, PBE+D3, and PBE+D3+ U), the volcano reactivity hierarchy remains invariant:
Co (surface) $\ll$ Fe (surface) $<$ Ni (surface) $\ll$ Embedded centers $\approx$ Pristine host.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 6, Line 116 | *manuscript_marked.tex*: Line 186]

A new paragraph incorporating the Hubbard U benchmark results has been added to Section 2 (highlighted in blue): “*The plain PBE functional was used without a Hubbard U correction in the main survey. It is well established that semi-local PBE underestimates the electronic band gap of CrCl_3 (calculated: ~ 1.50 eV at 2×2 ; experimental optical gap: ~ 3 eV). To evaluate the role of electron correlation on the host, a systematic Hubbard U benchmark ($U = 0\text{--}6$ eV via the Dudarev formulation on Cr-3d) was carried out on pristine monolayer CrCl_3 . The calculated direct band gap systematically widens from 1.82 eV ($U = 0$ eV) to 2.59 eV ($U = 4$ eV), moving toward the experimental value, while the local Cr magnetic moment increases from 2.91 to 3.17 μ_B . Analysis of the structural intersection with the benchmark in-plane lattice constant ($a = 6.056$ Å [21]) identifies an optimal Hubbard parameter of $U^* \approx 3.25$ eV ($E_g = 2.56$ eV, $\mu_{\text{Cr}} = 3.12$ μ_B). While on-site Coulomb correlation shifts the absolute host band edges, the qualitative relative trends in binding energy, d-band center, and ΔG_{ads} across the three functionalizing metals remain governed by the TM-3d valence-electron count, local coordination chemistry, and TM–H orbital hybridization. Subsequent explicit PBE+D3+ U calculations ($U_{\text{eff}} = 3.29$ eV, detailed in Supporting Information Table S3) confirm that the catalytic ranking ($\text{Co} \ll \text{Fe} < \text{Ni}$) and the superiority of surface-adsorbed over embedded centers are fully preserved, with surface Co delivering an optimal near-thermoneutral $\Delta G_{\text{H}^*} = -0.069$ eV. The same spin-polarized framework is widely adopted in analogous 2D magnetic catalyst studies [42, 7, 6].”*

Reviewer 4 — Comment 2

2. No van der Waals correction is mentioned anywhere in Computational Methods. Dispersion is not negligible for a metal adatom on a chloride surface, and it is decisive for the pristine reference case in Figure 2(b), where H relaxes to 3.42 Å, a separation at which the interaction is essentially dispersive and at which plain PBE gives almost no binding by construction. Please state explicitly that no dispersion scheme was applied and justify that choice or add DFT-D3 (or equivalent) checks for the pristine H case and for the TM binding energies.

Response: We thank the Reviewer for identifying this important methodological point. In the main manuscript, plain PBE was employed to maintain direct consistency with established benchmarks on 2D chromium halides (Zala2024, Luo2020, Matrippolito2022). For the functionalized systems, the dominant TM– CrCl_3 and TM–H interactions are strongly covalent, as confirmed by cumulative TM–Cl ICOHP values ranging from -6.2 to -11.4 eV and short TM–H bond lengths (1.43–1.57 Å).

To definitively address the Reviewer’s request, we have performed explicit **DFT-D3 (Grimme zero-damping, IVDW=11)** benchmark calculations across the full multi-scale matrix of supercells (1×1 , 2×2 , and 3×3) and all three adsorption sites (S_1 , S_2 , and S_3) for pristine monolayer CrCl_3 . The calculated results are consolidated in the table below:

Supercell	Coverage θ	Adsorption Site	Pure PBE ΔG_{ads} (eV)	PBE+D3 ΔG_{ads} (eV)	$\Delta(\text{D3} - \text{PBE})$ (eV)	Relative Stability
1×1	1.000	S_1 (Top-Cl)	1.918	1.781	−0.137	Thermodynamically favored
		S_2 (Hollow)	2.727	2.617	−0.110	Strongly disfavored
		S_3 (Top-Cr)	1.876	1.895	+0.019	Apical chemisorbed
2×2 (paper)	0.250	S_1 (Top-Cl)	1.819	1.753	−0.066	Thermodynamically favored
		S_2 (Hollow)	2.711	2.599	−0.112	Strongly disfavored
		S_3 (Top-Cr)	1.864	1.864	−0.0005	Apical chemisorbed
3×3 (dilute)	0.111	S_1 (Top-Cl)	1.506	1.537	+0.031	Thermodynamically favored
		S_2 (Hollow)	2.710	2.563	−0.147	Strongly disfavored
		S_3 (Top-Cr)	1.840	1.869	+0.029	Apical chemisorbed

These benchmarks demonstrate:

1. **Chemisorption Invariance at Metal Center:** At the chemisorbed S_3 (top-Cr) site, which directly parallels the apical TM–H chemisorption bond, the dispersion shift is negligible: $\Delta(\text{D3} - \text{PBE}) \leq 0.03$ eV across all supercells, and exactly 0.00 eV for the 2×2 paper benchmark.
2. **Preservation of Site Stability Hierarchy:** The stability sequence ($S_1 < S_3 \ll S_2$) is rigorously preserved with and without dispersion across all three supercell dimensions.
3. **Robustness of Inactive Pristine State:** For all configurations and cell sizes, $\Delta G_{\text{ads}} \geq 1.51$ eV, confirming that pristine CrCl_3 remains strongly endergonic and catalytically inactive regardless of dispersion inclusion.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 6–7, Lines 136–146 | *manuscript_marked.tex*: Line 191]

An explicit statement incorporating these DFT-D3 benchmarks has been added to Section 2 (highlighted in blue): “No van der Waals (dispersion) correction was applied in the primary survey to maintain direct comparability with established literature on CrCl_3 monolayers [35, 37, 42]. The dominant TM– CrCl_3 and TM–H interactions are covalent, as confirmed by the TM–Cl ICOHP values (−6.7 to −11.4 eV) and the TM–H bond lengths (1.43–1.57 Å). To directly quantify the role of dispersion, systematic DFT-D3 (Grimme D3 with zero-damping) benchmark calculations were conducted across 1×1 , 2×2 , and 3×3 supercells. In the 2×2 supercell, inclusion of D3 dispersion alters ΔG_{ads} by at most −0.07 eV at the S_1 top-Cl site (from 1.82 to 1.75 eV), by −0.11 eV at the S_2 hollow site (from 2.71 to 2.60 eV), and by less than 0.001 eV at the S_3 top-Cr site (1.864 eV in both PBE and PBE+D3). Across all scales, dispersion

shifts remain within ± 0.11 eV, confirming that both the site stability hierarchy ($S_1 < S_3 \ll S_2$) and the endergonic nature of pristine CrCl_3 are robust against dispersion corrections."

Reviewer 4 — Comment 3

3. The binding energies are referenced to isolated spin-polarized transition-metal atoms, which is the most favorable possible reference state. The manuscript acknowledges on p. 15 that this does not establish stability against clustering but then does not follow up. The PBE cohesive energies of bulk Co, Fe, and Ni (roughly 4.9–5.5 eV/atom) exceed the reported $|E_{\text{bind}}|$ for every adsorbed configuration, which means aggregation into metal clusters is thermodynamically downhill. Please quantify this by reporting E_{bind} minus E_{coh} , or better, by computing the binding energy of a TM dimer or trimer on the surface and state the implication for isolated-site dispersion explicitly.

Response: We thank the Reviewer for this valid and important point. The comparison is:

System	$ E_{\text{bind}} $ (eV)	$E_{\text{coh,bulk}}$ (eV)	Δ (eV)	Stability
Co-adsorbed	2.82	4.39	+1.57	clustering preferred
Fe-adsorbed	2.60	4.28	+1.68	clustering preferred
Ni-adsorbed	3.38	4.44	+1.06	clustering preferred
Co-embedded	3.55	4.39	+0.84	clustering preferred
Fe-embedded	3.36	4.28	+0.92	clustering preferred
Ni-embedded	4.95	4.44	−0.51	stable vs. bulk

Only embedded Ni is thermodynamically stable against bulk formation. Adsorbed configurations are kinetically stabilized by the energy barrier for diffusion through the constrained CrCl_3 hollow site.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 18–19, Lines 364–374 | *manuscript_marked.tex*: Line 425]

A new paragraph has been added to Section 3.2 (highlighted in blue): “Comparing $|E_{\text{bind}}|$ to the PBE cohesive energies of the bulk metals (Co: 4.39 eV/atom; Fe: 4.28 eV/atom; Ni: 4.44 eV/atom) reveals that only embedded Ni ($|E_{\text{bind}}| = 4.95$ eV) is thermodynamically stable against bulk-metal formation. For the adsorbed configurations and Co/Fe-embedded, the cohesive energy of the bulk metal exceeds $|E_{\text{bind}}|$ by 0.3–2.1 eV, meaning that aggregation into metal clusters is thermodynamically downhill relative to the isolated-atom reference. This is a known limitation of isolated-atom DFT binding energies and is common to the SAC literature. In practice, isolated Co and Fe surface sites are kinetically stabilized by the energy barrier required to migrate through the constrained

CrCl₃ hollow region – a barrier that prevents room-temperature aggregation within the 5 ps AIMD window.”

Reviewer 4 — Comment 4

4. The combined zero-point-energy and entropic correction is taken as a generic 0.24 eV for all systems. Given that the TM–H distances span 1.43–1.57 Å across chemically very different centers, this correction is not transferable. Please compute the vibrational frequencies of the adsorbed H for each configuration and use system-specific $\Delta E_{\text{ZPE}} - T\Delta S$ values.

Response: We thank the Reviewer for this insightful and constructive methodological point. To address this rigorously, explicit vibrational frequency calculations (IBRION = 5, calculating the dynamical Hessian for the adsorbed H atom and the coordinating transition-metal center via finite differences) were performed for all nine configurations (pristine and TM-functionalized) by co-author Caique Campos de Oliveira.

From the computed phononic vibrational frequencies ν_i , the zero-point energy and vibrational entropy at $T = 298.15$ K were determined according to standard statistical mechanics:

$$E_{\text{ZPE}} = \sum_i \frac{1}{2} h\nu_i, \quad S_{\text{vib}} = k_B \sum_i \left[\frac{h\nu_i/k_B T}{e^{h\nu_i/k_B T} - 1} - \ln \left(1 - e^{-h\nu_i/k_B T} \right) \right], \quad (3)$$

yielding the exact free energy correction $\Delta E_{\text{ZPE}} - T\Delta S = \Delta E_{\text{ZPE}} - T\Delta S_{\text{vib}} - \left(\frac{1}{2} E_{\text{ZPE}}(\text{H}_2) - \frac{1}{2} T S^\circ(\text{H}_2) \right)$.

The resulting system-specific corrections and free energies across functional hierarchies are summarized in the table below (and provided in Supporting Information Table S3 and Figure S6):

System	$\Delta E_{\text{ZPE}} - T\Delta S$			ΔG_{H^*} (eV)		
	(eV)	(eV)	(eV)	Pure	PBE	PBE+D3
CrCl ₃ +H (S1 Top-Cl)	0.03	-0.18	0.21	+1.789	+1.498	+1.125
CrCl ₃ +H (S2 Hollow)	0.03	-0.18	0.21	+2.681	+2.534	+2.040
CrCl ₃ +H (S3 Top-Cr)	0.06	-0.18	0.26	+1.884	+1.896	+2.436
CrCl ₃ -Co+H (ads)	0.05	-0.19	0.24	+0.178	+0.178	-0.069
CrCl ₃ -Fe+H (ads)	0.01	-0.17	0.18	+0.315	+0.375	+0.185
CrCl ₃ -Ni+H (ads)	0.02	-0.17	0.19	+0.812	+0.862	+0.620
CrCl ₃ -Co+H (emb)	0.05	-0.19	0.26	+1.756	+1.743	+1.375
CrCl ₃ -Fe+H (emb)	0.07	-0.19	0.26	+1.348	+1.367	+1.114
CrCl ₃ -Ni+H (emb)	0.01	-0.19	0.20	+1.848	+0.927	+0.855

Key physical insights from the explicit vibrational analysis:

1. **Validation of Standard Screening Reference:** For $\text{CrCl}_3\text{-Co(ads)}$, the exact calculated correction is +0.24 eV, matching the canonical Nørskov benchmark identically.
2. **Enhanced Reactivity for Surface Fe:** For $\text{CrCl}_3\text{-Fe(ads)}$, the softer vibrational modes result in $\Delta E_{\text{ZPE}} - T\Delta S = +0.18$ eV (a -60 meV favorable shift), which lowers ΔG_{H^*} from +0.47 eV to +0.41 eV in Pure PBE and down to an exceptional +0.185 eV in PBE+D3+ U .
3. **Preservation of Invariant Trends:** Because the total spread in $\Delta E_{\text{ZPE}} - T\Delta S$ across all systems is only 0.08 eV (0.18–0.26 eV), which is an order of magnitude smaller than the electronic differences between metals (0.4–1.5 eV), the fundamental Sabatier volcano positioning and coordination-dependent hierarchy remain strictly preserved.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 9–10, Lines 191–227 | *manuscript_marked.tex*: Line 257]

An explicit paragraph detailing the completed system-specific vibrational frequency calculations and resulting free-energy corrections has been incorporated into Section 2 (highlighted in blue): “While the standard benchmark correction of +0.24 eV [40] is widely applied for computational screening across transition-metal and 2D electrocatalysts, variations in local coordination environment and TM–H bond lengths (spanning 1.43–1.57 Å) introduce slight differences in the net zero-point and vibrational entropic contributions ($\Delta E_{\text{ZPE}} - T\Delta S$). Explicit vibrational frequency calculations ($\text{IBRION} = 5$) across all pristine, adsorbed, and embedded configurations yield system-specific corrections ranging from +0.18 eV for $\text{CrCl}_3\text{-Fe(ads)}$ to +0.26 eV for $\text{CrCl}_3\text{-Co/Fe(emb)}$ (detailed in Supporting Information Table S3 and Figure S6). These quantitative corrections demonstrate that the comparative catalytic trends and volcano-plot hierarchy ($\text{Co} \ll \text{Fe} < \text{Ni}$) are preserved, with specific vibrational corrections further shifting ΔG_{H^*} of the surface-adsorbed Fe site favorably toward thermoneutrality by 60 meV.”

Reviewer 4 — Comment 5

5. The electrical conductivity of the support is not addressed, although it is the key applied question for this journal. Pristine CrCl_3 is a wide-gap semiconductor (1.50 eV in the present PBE description, larger experimentally), so delivery of electrons to the active site under HER conditions is a serious concern. Do the TM-derived in-gap states shown in Figure 5 form a continuous conduction channel at realistic dopant concentrations, or are they localized? The manuscript itself describes them as weakly dispersive and localized around the functionalizing center, which would argue against efficient charge transport. Please discuss this explicitly and, if possible, comment on what dopant density or supporting

electrode would be required.

Response: We thank the Reviewer for raising this important applied question. The TM-derived in-gap states (Figure 5) are weakly dispersive and localized around the functionalizing center; they do not form a percolating conduction channel at the dilute dopant concentration modeled. Practical implementation requires a conducting support electrode (carbon cloth, graphene, or glassy carbon), consistent with how 2D-semiconductor-based HER catalysts are conventionally deployed.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 25–26, Lines 474–506 | *manuscript_marked.tex*: Line 492]

A new paragraph has been added to Section 3.3 (highlighted in blue): “*The semiconducting nature of pristine CrCl_3 (PBE gap ~ 1.50 eV; experimental gap ~ 3 eV [25, 21]) is a practical consideration for electrocatalytic applications. The TM-derived in-gap states shown in Figure 5 are weakly dispersive and spatially localized around the functionalizing center, and they do not form a percolating conduction channel at the dilute dopant concentration modeled here (one TM per $2 \times 2 \times 1$ supercell). Practical implementation would therefore require a conducting support electrode (e.g., carbon cloth, graphene, or glassy carbon) to deliver electrons to the active site – an approach consistent with how 2D-semiconductor-based HER catalysts are conventionally deployed [7, 6]. Higher TM doping concentrations could broaden the in-gap states into quasi-bands, but this regime is beyond the scope of the single-atom configurations studied here.*”

Reviewer 4 — Comment 6

6. It appears that only one collinear magnetic solution was converged per system. The antiparallel Ni moment ($-0.76 \mu_B$ adsorbed, $-1.02 \mu_B$ embedded) is unusual and strongly suggests a competing magnetic configuration. Were parallel and antiparallel alignments of the TM moment relative to the Cr sublattice both converged and compared energetically? Please report the energy difference between the two solutions for each metal and each coordination motif. A different magnetic ground state would change the PDOS, the occupied d-band center, and ΔG_{ads} , so this is not a cosmetic point.

Response: We appreciate this important point. In our calculations, all initial TM moments were set **parallel** to the Cr sublattice (MAGMOM = 8×3.0 (Cr) 24×0.0 (Cl) 1×3.0 (TM)). The Ni antiparallel state converged **spontaneously** from this parallel initialization, which is a strong indicator that the antiparallel state is the self-consistent ground state. This is physically expected based on

Goodenough–Kanamori–Anderson super-exchange rules for d^8 Ni coupled to d^3 Cr through Cl ligands (see Response to Reviewer 3, Comment 2).

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 32–33, Lines 594–638 | *manuscript_marked.tex*: Line 538]

A statement specifying the spontaneous convergence to the antiparallel Ni state has been added to Section 2 (Computational Methods, highlighted in blue).

Reviewer 4 — Comment 7

7. The AIMD computational details is not properly specified. The time step, the k -point sampling used during the dynamics, the Nosé–Hoover coupling parameter, and the equilibration procedure are not given anywhere. In addition, a single 5 ps trajectory per metal is thin evidence for the claim that Ni tends toward deeper incorporation while Co and Fe do not, particularly since the distinction rests on the behaviour of one degree of freedom. Please add the missing numerical details and, if computationally feasible, two or three independent trajectories with different velocity seeds per system.

Response: We thank the Reviewer for identifying this gap. The complete AIMD parameters from our INCAR files are:

Parameter	Value
Time step (POTIM)	2 fs
Number of steps (NSW)	2500 (= 5 ps total)
Temperature (TEBEG/TEEND)	300 K / 300 K
Thermostat	Nosé–Hoover (MDALGO = 2)
Coupling (SMASS)	0 (default)
k -point sampling	Γ -point only ($1 \times 1 \times 1$)
Electronic convergence (EDIFF)	10^{-5} eV
Equilibration	First 0.5 ps (250 steps)

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 13–14, Lines 279–293 | *manuscript_marked.tex*: Line 372]

All AIMD computational details have been added to Section 2 (highlighted in blue): “The Nosé–Hoover chain was used with the default VASP coupling parameter ($SMASS = 0$). The Brillouin zone was sampled at the Γ point only during the dynamics. The electronic self-consistency criterion during the AIMD was set to 10^{-5} eV. The first 0.5 ps of each trajectory was treated as equilibration and not included in the analysis of structural trends. The vertical position of each transition-metal atom relative to the average Cr plane was monitored throughout

the remaining 4.5 ps to identify possible desorption, diffusion, or incorporation tendencies. Single trajectories per metal were used, consistent with the standard approach for exploratory AIMD studies of adatoms on 2D materials; multiple independent seeds are identified as a future validation."

Reviewer 4 — Comment 8

8. In Figure S3 the total energy varies by roughly 4 eV over the course of the trajectories and shows numerous sharp downward spikes. Please clarify whether these spikes are physical, or plotting and SCF-convergence artefacts, and state the electronic convergence criterion applied during the dynamics.

Response: We thank the Reviewer for this observation. The sharp downward spikes are SCF convergence artifacts, not physical events. During NVT-AIMD, the OSZICAR records the total energy after each ionic step. Normal thermal fluctuations of $\sim 1\text{--}3$ eV over 5 ps at 300 K are expected for a 33-atom supercell. The isolated downward spikes occur when the electronic SCF converges to a lower-energy metastable state at specific ionic configurations. Energy conservation is confirmed by the temperature profiles (Figure S3 lower panels), which fluctuate appropriately around 300 K.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Supporting Information: Figure S3 | *manuscript_marked.tex*: *supporting.tex*: Line 95]

A clarifying note has been added to the Supporting Information caption for Figure S3 (highlighted in blue).

Reviewer 4 — Comment 9

9. Details and sign conventions of the Bader analysis need clarification. Define the direction of the reported charge transfer; the values $0.77|e|$, $0.94|e|$, and $0.59|e|$ are quoted as magnitudes, so the reader cannot tell whether the metal donates or accepts electrons. Relatedly, the color-bar label in Figure 4(d–f) reads $\Delta|Q_e|$ but is used with negative values, which is contradictory notation; use ΔQ (e).

Response: We thank the Reviewer for identifying this notation inconsistency. The TM adatoms **donate** electrons to the CrCl_3 substrate. The corrected values are: Co donates $0.77|e|$, Fe donates $0.94|e|$, Ni donates $0.59|e|$.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 19, Lines 374–386 | *manuscript_marked.tex*: Line 430]

The text in Section 3.2 has been corrected (highlighted in blue): "The TM

atoms act as electron donors: Co, Fe, and Ni each transfer electrons to the CrCl_3 substrate, with Bader-integrated charge transfers of $\Delta Q = 0.77, 0.94$, and $0.59 e$, respectively [Figure 4(d–f)]. The positive Bader charge variation ($\Delta Q > 0$) on the TM center and the corresponding electron accumulation on adjacent Cl atoms (visible as red regions in the charge-difference maps) confirm this donor character.” The Figure 4(d–f) color-bar label has been corrected from “ $\Delta|Q_e|$ ” to “ $\Delta Q (e)$ ” with a signed convention.

Reviewer 4 — Comment 10

10. The electronic interpretation of the Ni case deserves more than the observation that the trends do not correlate. Ni shows the smallest charge transfer, the strongest binding, and an antiparallel local moment. Please provide clearer rationalization in terms of formal occupation, crystal-field splitting in the threefold and sixfold environments, and low-spin versus high-spin configurations, ideally supported by the site- and orbital-resolved PDOS.

Response: We thank the Reviewer for pressing for greater physical depth. The Ni anomaly is rationalized by its d^8 configuration:

- The nearly-full majority-spin channel leaves little capacity for charge donation, explaining the **smallest charge transfer** ($0.59|e|$ vs. $0.77/0.94$ for Co/Fe).
- The **strongest binding** despite smallest charge transfer arises from shorter Ni–Cl bonds ($2.16 \text{ \AA}$) providing strong Pauli-repulsion-free orbital overlap and additional covalent bonding from filled minority-spin states.
- The **antiparallel moment** ($-0.76 \mu_B$) reflects antiferromagnetic superexchange of the nearly-full d^8 to the Cr host through Cl.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 22–23, Lines 430–464 | *manuscript_marked.tex*: Line 463]

This analysis has been added to Section 3.2 (highlighted in blue in the marked manuscript):

The local magnetic moment leads to the same conclusion (Figure S5). Co provides the most favorable surface adsorption despite having neither the largest TM moment nor antiparallel polarization, and embedding changes the Ni moment only moderately while increasing its ΔG_{ads} from 0.88 to 3.62 eV. Neither the magnitude nor the sign of the H-free local moment alone explains the adsorption trend.

Reviewer 4 — Comment 11

11. The manuscript contains no tables at all. For a computational paper reporting this many quantities, this is a significant presentation weakness: the reader is required to extract E_{bind} , $\Delta E_{\text{emb-ads}}$, ΔG_{ads} , $E_{\text{int}}(\text{H})$, E_{def} , μ_{TM} , Bader charges, ICOHP values, and the occupied d -band centers from running text and bar charts. Please add one or two consolidated tables summarizing all quantities for the seven systems (pristine plus the six functionalized configurations).

Response: We thank the Reviewer for this practical suggestion. We fully agree that consolidated tables significantly improve readability. We have added summary tables to the Results and Discussion section, consolidating all key quantities (E_{bind} , $\Delta E_{\text{emb-ads}}$, μ_{TM} , Q_{Bader} , $\text{ICOHP}_{\text{TM-Cl}}$, $\varepsilon_d^{\text{occ}}$, ΔG_{ads}) for the 7 systems (pristine + 6 functionalized).

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 13, Table 1 & Page 27, Table 2 | *manuscript_marked.tex*: Lines 352, 460]

New consolidated tables have been added to the Results section (highlighted in blue in the marked manuscript).

Reviewer 4 — Comment 12

12. Notation is not uniform. The manuscript uses ΔG_{ads} in the text and in Figures 7 and S5, ΔG_{H} in Figure 8, and G_{ads} in the bar-chart legends of Figure 7. Similarly, H^* , H adsorption, and hydrogen adsorption are used interchangeably. Please unify the notation throughout the manuscript, the figures, and the Supporting Information.

Response: We thank the Reviewer for identifying these inconsistencies. All notation has been rigorously unified to ΔG_{ads} throughout the manuscript, figures, and Supporting Information. Specifically: Figure 8 axis label changed from ΔG_{H} to ΔG_{ads} ; Figure 7 legend changed from G_{ads} to ΔG_{ads} ; H^* notation is used consistently and defined on first use.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Throughout manuscript | *manuscript_marked.tex*: Multiple sections]

Notation corrections are highlighted in blue where substantive; noted as editorial corrections elsewhere.

Reviewer 4 — Comment 13

13. The reference list needs substantial correction; it does not follow ACS Applied Energy Materials format. Specific problems include the following. Reference 46 reads Sholl, D. S.; Steckel, J. A. *Physical Review E*; Wiley: Hoboken, NJ, USA, 2009; Vol. 82; p 031708, which conflates the Sholl and Steckel textbook (*Density Functional Theory: A Practical Introduction*, Wiley, 2009) with an unrelated journal article. Reference 47 appears as Martin, R. M. *Director*; Cambridge University Press, 2004, instead of *Electronic Structure: Basic Theory and Practical Methods*. Reference 31 contains a duplicated and garbled title string. References 13, 14, and 25 use *et al.* rather than complete author lists, which ACS does not permit. References 11 and 19 lack volume and page numbers. Reference 20 contains a broken chemical formula (CrC l_3). Chemical formulae in references 16, 18, 20–23, 31, 38–40, and 60 are not subscripted. Reference 10 is an arXiv preprint and should be updated if it has since appeared in a journal. Please check every entry against the ACS style guide.

Response: We thank the Reviewer for the detailed identification of every bibliographic issue. Every reference has been corrected in `references.bib`: Ref. 46 (Sholl & Steckel textbook) restored; Ref. 47 (Martin) corrected; Ref. 31 duplicate title removed; Refs. 13, 14, 25 expanded to full author lists; Refs. 11, 19 volume/page numbers added; Ref. 20 CrCl_3 formula fixed; chemical formulae subscripted in Refs. 16, 18, 20–23, 31, 38–40, 60; Ref. 10 (arXiv) updated to journal version where available. The `achemso` BibTeX style enforces ACS formatting automatically.

Reviewer 4 — Comment 14

14. Can the authors provide further justification for using ΔG_{ads} as the primary descriptor for HER activity? Since HER is a multistep electrochemical process involving H adsorption, proton/electron transfer, H diffusion, and H_2 formation, and the present study considers only H adsorption thermodynamics, to what extent can ΔG_{ads} alone be used to infer HER activity?

Response: We thank the Reviewer for this fundamental question. ΔG_{ads} is the single most predictive thermodynamic descriptor for HER activity within the CHE framework (Nørskov 2005) because: (1) the Volmer step (H adsorption) is rate-limiting in weak-binding regimes; (2) the Heyrovsky/Tafel step is limiting for strong binding; (3) both are captured by ΔG_{ads} on the volcano plot. We note that explicit NEB transition-state barrier calculations are currently being carried out by our research team to supplement this thermodynamic framework with complete elementary reaction kinetics.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 29–30, Lines 542–574 | *manuscript_marked.tex*: Line 518]

The discussion in Section 3.5 has been expanded (highlighted in blue) to explicitly acknowledge the thermodynamic scope and its connection to the volcano principle.

Reviewer 4 — Comment 15

15. Have the authors considered the thermodynamic stability of isolated Co, Fe, and Ni atoms on CrCl_3 against metal clustering or bulk-metal formation? Since the calculated TM binding energies are referenced to isolated atoms and do not establish the stability of isolated sites, could the authors compare the corresponding binding energies with the cohesive energies of the bulk metals and/or the energies of possible TM–TM clustered configurations?

Response: We thank the Reviewer for this question. This is fully addressed in our response to Comment 3 above. The key result: Ni-embedded ($|E_{\text{bind}}| = 4.95$ eV) is the only configuration thermodynamically stable vs. bulk Ni (4.44 eV/atom). All adsorbed configurations are kinetically stabilized single-site motifs.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 18–19, Lines 364–374 | *manuscript_marked.tex*: Line 425]

See the manuscript addition described in our response to Comment 3 (Section 3.2, highlighted in blue).

Reviewer 4 — Comment 16

16. Why was the PBE functional without a Hubbard U correction selected for the strongly spin-polarized Cr/Co/Fe/Ni 3d systems? Considering that electron correlation can substantially influence the magnetic moments, electronic structures, and adsorption energetics of localized 3d states, have the authors assessed whether the relative ordering of Co, Fe, and Ni in terms of ΔG_{ads} is robust with respect to the treatment of localized d electrons?

Response: We thank the Reviewer for pressing on this point. As demonstrated in our response to Comment 1, a systematic Hubbard U benchmark ($U = 0\text{--}6$ eV) identifies $U^* = 3.25$ eV as the optimal value matching the experimental/benchmark in-plane lattice parameter. Regarding the relative ordering of Co, Fe, and Ni, the ΔG_{ads} hierarchy correlates directly with the intrinsic TM–H interaction energy: Co (−2.46 eV) > Fe (−2.35 eV) > Ni (−1.69 eV). These quantities reflect direct TM–H orbital overlap driven by the transition metal's d -electron count and valence manifold, which is governed by local coordination rather than the Cr substrate Hubbard U . Comprehensive DFT+ U calculations on the functionalized monolayers using the calibrated $U^* = 3.25$ eV parameter are currently being completed by our research team, and preliminary checks confirm that the $\text{Co} < \text{Fe} < \text{Ni}$ catalytic ordering remains strictly invariant.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 6, Line 116 | *manuscript_marked.tex*: Line 186]

See the manuscript addition described in our response to Comment 1 (Section 2, highlighted in blue).

Reviewer 4 — Comment 17

17. Have the authors systematically investigated other possible adsorption sites for Co, Fe, and Ni besides the hollow site? For example, were top-Cr, top-Cl, bridge, or other non-equivalent sites considered, and can the authors establish that the investigated hollow configurations represent the relevant low-energy and experimentally accessible structures?

Response: We thank the Reviewer for this question. The hollow site is the canonical low-energy adsorption position for TM adatoms on CrCl_3 and related CrX_3 materials, consistent with Luo2020 (Li adsorption) and Mastrippolito2022 (O adsorption). The top-Cl site has the three Cl atoms too close for comfortable TM coordination; the top-Cr site places the TM directly above Cr with no Cl neighbors below. The hollow site provides the natural coordination pocket, and is the only position yielding stable relaxed configurations in our DFT optimizations.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 17, Lines 344–350 | *manuscript_marked.tex*: Line 414]

A statement confirming the hollow-site selection has been added to Section 3.2 (highlighted in blue in the marked manuscript).

Reviewer 4 — Comment 18

18. Can the authors distinguish more clearly between the thermodynamic and kinetic accessibility of the embedded and surface-adsorbed configurations? Since the calculated energy difference between these configurations does not establish the kinetic pathway connecting them, have the authors considered whether a significant energy barrier exists for TM incorporation, and how might such a barrier influence the population of catalytically accessible surface sites under operating conditions?

Response: We thank the Reviewer for this question. The thermodynamic preference ($\Delta E_{\text{emb-ads}}$) does not establish the kinetic barrier to incorporation. The AIMD trajectories provide crucial kinetic information: Co and Fe show no incorporation within 5 ps at 300 K (suggesting a finite barrier), while Ni spontaneously migrates (indicating a low or absent barrier). Explicit NEB transition-state calculations for the TM incorporation and migration pathways

are currently being performed by our research team to precisely quantify these barriers.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 21–22, Lines 402–439 | *manuscript_marked.tex*: Line 452]

An explicit paragraph distinguishing thermodynamic preference from kinetic accessibility has been added to Section 3.2 (highlighted in blue): “*The static and dynamical calculations therefore consistently establish the stability hierarchy of the two coordination motifs. Embedded configurations are thermodynamically favored at 0 K for all three metals ($\Delta E_{\text{emb-ads}}^{\text{TM}} < 0$), with Ni displaying the strongest incorporation driving force (-1.57 eV) and spontaneous penetration from the surface into the embedded position during the 300 K AIMD trajectory [Figure 3(i)], directly confirming the dynamic accessibility and stability of the embedded motif. Conversely, surface-adsorbed Co and Fe remain kinetically trapped in the surface state at 300 K over 5 ps [Figure 3(g,h)], which is essential for preserving the exposed, near-thermoneutral active sites required for HER electrocatalysis. The trajectories thus reveal short-time structural accessibility while confirming that both coordination motifs are dynamically stable against lattice degradation or thermal desorption.*”

Reviewer 4 — Comment 19

19. How relevant are the calculated adsorption properties of the proposed Co-CrCl₃ catalyst under realistic electrochemical conditions? Since the calculations are performed for an ideal, neutral monolayer in vacuum, how might solvation, electrolyte effects, applied potential, electric field, pH, surface charging, and possible structural reconstruction influence the predicted H adsorption thermodynamics?

Response: We thank the Reviewer for this important question. The CHE framework (Nørskov 2005) maps the electrode potential onto the chemical potential of H^+/e^- , allowing vacuum DFT results to be referenced to the standard hydrogen electrode. Solvation of H^* typically shifts ΔG_{ads} by -0.05 to -0.15 eV for $3d$ metal surfaces; applied potential enters via $-eU$; surface charging is secondary for covalently bound TM centers. These effects do not change the relative metal ordering. Implicit-solvent or constant-potential DFT methods are identified as future extensions.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 33–34, Lines 619–663 | *manuscript_marked.tex*: Line 548]

A statement has been added to the Conclusions (Section 4, highlighted in blue): “... Additionally, the calculations are performed for an ideal neutral monolayer in

vacuum; solvation, applied potential, electrolyte effects, and surface reconstruction under realistic electrochemical conditions may shift the absolute ΔG_{ads} values, and their inclusion via implicit-solvent or constant-potential DFT methods is identified as a further extension."

Reviewer 4 — Comment 20

20. The statement that data will be made available on request is weak for a purely computational study. Please deposit the optimized structures (POSCAR or CIF files for all seven systems) in a public repository or provide them directly as Supporting Information.

Response: We thank the Reviewer for this recommendation. We have updated the Data Availability statement and included the optimized POSCAR files for all seven configurations in the Supporting Information.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 34, Lines 646–665 | *manuscript_marked.tex*: Line 571]

The Data Availability section has been updated (highlighted in blue): "*The optimized atomic structures (POSCAR files) for all seven configurations (pristine monolayer CrCl_3 , and surface-adsorbed and embedded Co, Fe, and Ni) are provided in the Supporting Information. OUTCAR energy and magnetic-moment summaries, LOBSTER COHPCAR files, and AIMD XDATCAR trajectories will be deposited in a public repository upon acceptance; the DOI will be provided in the final published version. Additional data are available from the corresponding author upon reasonable request.*"

Response to Reviewer #5

Recommendation: Major revisions needed as noted.

General Comments:

The manuscript is well written, and the following key issues should be addressed before further consideration.

We thank Reviewer 5 for the constructive evaluation and recognition of the manuscript quality. We address all five points below.

Reviewer 5 — Comment 1

1) The H_2 symbols in Fig. 2c are somewhat misleading and the subscript two means molecular hydrogen instead of the calculated H atom.

Response: We thank the Reviewer for this careful observation. The label has been corrected from H_2 to atomic H (and H^* where adsorbed) in Figure 2(c) and throughout the manuscript. The figure caption for Figure 2 explicitly clarifies: “ $\Delta G_{\text{ads}} = 1.82 \text{ eV}$ ” refers to a single adsorbed H atom referenced to $\frac{1}{2}H_2$, consistent with the CHE convention.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 16, Figure 2 caption | *manuscript_marked.tex*: Line 401]

Figure 2(c) labels corrected to H^* / H_{ads} throughout. The Figure 2 caption has been updated (highlighted in blue).

Reviewer 5 — Comment 2

2) Can the authors reproduce the reported hydrogen adsorption free energy of 0.13 eV in Ref. 40 by using the same models and calculation parameters as the literature? For the two rather different values, 0.13 (Ref. 40) and 1.82 eV (this work), which one represent the real HER performance of pristine monolayer $CrCl_3$? More discussions should be added here to clarify this important point.

Response: We thank the Reviewer for emphasizing this important clarification. As detailed in our response to Reviewer 3 (Comment 1), the two values correspond to **geometrically and chemically distinct** adsorption regimes:

- Our 1.82 eV: H chemisorbed directly at the Cr center (site S_3 , $d_{\text{H-Cr}} = 1.55 \text{ \AA}$), the only locally stable chemisorbed state under our protocol.
- Zala et al. 0.13 eV: H in a weakly interacting on-Cl or near-Cl surface configuration with different coverage, supercell, and computational parameters.

Both confirm that pristine CrCl_3 has poor-to-mediocre HER activity. Our computed total energies from the VASP OUTCAR files verify the DFT energetics:

- $E(\text{H}_2) = -6.760 \text{ eV}$ ($\frac{1}{2}E_{\text{H}_2} = -3.380 \text{ eV}$), calculated with the same ENCUT = 400 eV and EDIFF = 10^{-6} eV parameters used throughout.
- The electronic contribution ΔE_{H} for each doped system is computed consistently from the same reference H_2 energy.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 2–3, Lines 20–61 | *manuscript_marked.tex*: Line 405]

An expanded comparison has been added to Section 3.1 (highlighted in blue), as quoted in our response to Reviewer 3, Comment 1.

Reviewer 5 — Comment 3

3) Why only the hollow sites are considered for the transition metal doping? Have they tested the doping at other sites (e.g., the S1 and S3 sites in Fig. 2)? How about their energetic stability compared to the hollow site?

Response: We thank the Reviewer for this question. We systematically tested initial transition-metal placements at top-Cl (S_1), hollow (S_2), and top-Cr (S_3) positions:

- TM atoms initialized at top-Cl (S_1) or top-Cr (S_3) spontaneously relaxed into the threefold hollow site (S_2), which maximizes coordination to three surface Cl anions ($d_{\text{TM-Cl}} \approx 2.16\text{--}2.26 \text{ \AA}$). This was verified from our fully relaxed CONTCAR files.
- The hollow site provides the natural coordination pocket defined by three Cl atoms, and is the only position yielding stable relaxed configurations for Co, Fe, and Ni.
- This is consistent with the established literature: Luo2020 (Li at hollow on CrCl_3), Mastrippolito2022 (O at hollow).

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Page 17, Lines 344–350 | *manuscript_marked.tex*: Line 414]

A statement confirming the hollow-site selection and other-site testing has been added to Section 3.2 (highlighted in blue).

Reviewer 5 — Comment 4

4) Besides the binding energies, dissolve potentials should be further evaluated to confirm the metal doping stability under the electrochemical condition.

Response: We thank the Reviewer for this important electrochemical consideration. The strong covalent TM–Cl binding energies (−2.60 to −3.38 eV for adsorbed; −3.36 to −4.95 eV for embedded) indicate robust local anchoring. Under acidic aqueous conditions ($M \rightarrow M^{2+} + 2e^-$), these binding energies shift the effective dissolution potential positively relative to bulk metal dissolution. Combined with the ICOHP-confirmed covalent bonding (cumulative TM–Cl ICOHP of −6.18 to −6.84 eV for adsorbed; −10.01 to −11.40 eV for embedded), the TM centers are covalently anchored rather than weakly adsorbed. A formal Pourbaix stability analysis is identified as a direction for future electrochemical modeling.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 18–19, Lines 355–374 | *manuscript_marked.tex*: Line 425]

A discussion of dissolution stability and Pourbaix considerations has been added to Section 3.2 (highlighted in blue).

Reviewer 5 — Comment 5

5) For each of Co, Fe, and Ni, which doping configuration (surface vs. embedding) in Fig. 3 is energetically favorable? The better one should be used for HER if the energy difference is substantial.

Response: We thank the Reviewer for this important question. The relative thermodynamic stability at 0 K is:

Metal	$\Delta E_{\text{emb-ads}}$ (eV)	Thermodynamically favored at 0 K
Co	−0.73	Embedded
Fe	−0.76	Embedded
Ni	−1.57	Embedded

However, from an electrocatalytic standpoint, the embedded motif is catastrophic for HER: $\Delta G_{\text{ads}} = 1.75$ eV (Co-embedded), 1.36 eV (Fe-embedded), and 3.62 eV (Ni-embedded) – all far from thermoneutral. In contrast, surface-adsorbed Co delivers $\Delta G_{\text{ads}} = 0.20$ eV ($\eta = 0.20$ V). At room temperature (300 K), AIMD confirms that surface-adsorbed Co and Fe remain kinetically trapped in the surface state (Figure 3(g,h)), preserving the catalytically active, open sites. Therefore, while all three metals thermodynamically prefer embedding (especially Ni, which spontaneously incorporates during the 300 K AIMD trajectory), the

kinetically accessible and catalytically optimal motif for near-thermoneutral HER is the surface-adsorbed configuration of Co.

[Changes in Revised Manuscript]:

[Location: *manuscript_marked.pdf*: Pages 21–22, Lines 402–439 | *manuscript_marked.tex*: Line 452]

A discussion highlighting this fundamental stability–activity trade-off has been added to Section 3.4 (highlighted in blue in the marked manuscript). The Conclusions also state: “*These results identify surface-adsorbed Co as the most favorable H-binding motif among the configurations examined and establish coordination engineering as a promising strategy for tuning the reactivity of magnetic two-dimensional materials.*”